9.4: Fredholm Alternative and Generalized Green's Functions



9.5: Green's Functions for Poisson's Equation

James Rosado

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9.4: Fredholm Alternative and Generalized Green's Functions

Recall the steady-state heat equation with homogeneous boundary conditions with time-independent source *Q*(*x*):

$$0 = k\frac{d^2u}{dx^2} + Q(x)$$

Boundary Conditions: $u(0) = u(L) = 0$

For simplicity we will let $f(x) = -\frac{Q(x)}{k}$ and we will let the operator $\mathrm{L}[u] = \frac{d^2}{dx^2}(u)$. This will reduce our differential equation to the form:

$$\mathrm{L}[u] = f(x)$$

9.4: Fredholm Alternative and Generalized Green's Functions

But, remember we can express our solution as an infinite series expansion using the eigen -functions we obtain from the eigenvalues λ:

$$u(x) = \sum_{n=1}^{\infty} a_n \phi_n(x)$$

Recall that $\phi_n(x) = \sin\left(\frac{n\pi}{L}x\right)$ and $\phi_n''(x) = -\lambda_n \phi_n(x)$ where $\lambda_n = \left(\frac{n\pi}{L}\right)^2$. So our differential equation becomes:

9.4: Fredholm Alternative and Generalized Green's Functions

The last equation has the following implication:

$$-\sum_{n=1}^{\infty} a_n \lambda_n \phi_n(x) = f(x) \quad \rightarrow \quad -a_n \lambda_n = \frac{\int_0^L f(x)\phi_n(x)\,dx}{\int_0^L \phi_n^2(x)\,dx}$$

If we substitute back into our solution u(x), we get:

$$u(x) = \sum_{n=1}^{\infty} a_n \phi_n(x) = \sum_{n=1}^{\infty} \left(\frac{\int_0^L f(x_0)\phi_n(x_0)\,dx_0}{-\lambda_n \int_0^L \phi_n^2(x)\,dx} \right) \phi_n(x)$$

9.4: Fredholm Alternative and Generalized Green's Functions

$$\sum_{n=1}^{\infty}\left(\frac{\int_0^L f(x_0)\phi_n(x_0)\,dx_0}{-\lambda_n\int_0^L \phi_n^2(x)\,dx}\right)\phi_n(x)$$

Notice that the denominator is just going to be a constant because it only depends on one variable. We can bring the top integral outside of the summation to get,

$$u(x)=\int_0^L f(x_0)\sum_{n=1}^{\infty}\frac{\phi_n(x)\phi_n(x_0)}{-\lambda_n\int_0^L \phi_n^2(x)\,dx}\,dx_0=\int_0^L f(x_0)G(x,x_0)\,dx_0$$

This solution works ideally if $\lambda_n \neq 0$. But what happens if this is not the case?

9.4: Fredholm Alternative and Generalized Green's Functions

So what do we do when λ = 0? Recall,

$$\mathrm{L}[u] = \sum_{n=1}^{\infty} a_n \frac{d^2\phi_n}{dx^2} = -\sum_{n=1}^{\infty} a_n \lambda_n \phi_n(x)$$

Or simply,

$$\mathrm{L}[\phi_n(x)] = -\lambda_n \phi_n(x)$$

The statement above means that each eigen-function ϕ(x) is a solution to the differential equation with the same homogeneous boundary conditions.

9.4: Fredholm Alternative and Generalized Green's Functions

Let us suppose that λ = 0 is one of the eigenvalues and has a corresponding eigen-function $\phi_h(x)$. Then,

$$\mathrm{L}[\phi_n(x)] = -\lambda_n \phi_n(x) \;\rightarrow\; \mathrm{L}[\phi_h(x)] = 0$$

This means that $\phi_h(x)$ is non-trivial; therefore, nonhomogeneous solutions solving the same homogeneous B.C. conditions are equivalent to eigen-functions that correspond to the zero eigenvalue.

If there are no NON-TRIVIAL homogeneous solutions then λ = 0 is not one of the eigenvalues.

9.4: Fredholm Alternative and Generalized Green's Functions

Okay so what does that mean?? Here is an example, suppose we are given the following:

$$u_{xx} + u = e^x \quad \text{with} \quad u(0) = u(\pi) = 0$$

The corresponding homogeneous time independent equation is:

$$\phi''(x) + \phi(x) = -\lambda\phi(x) \quad \text{with} \quad \phi(0) = \phi(\pi) = 0$$

Bring terms to one side yields: $\phi'' + (\lambda + 1)\phi = 0$, which has characteristic equation,

$$r^2 + (\lambda + 1) = 0$$

If we solve we get: $r_{1,2} = \pm i\sqrt{\lambda + 1}$

9.4: Fredholm Alternative and Generalized Green's Functions

With pure imaginary roots this means our solution will be of the form:

$$\phi(x) = A\cos\left(\sqrt{\lambda+1}\cdot x\right) + B\sin\left(\sqrt{\lambda+1}\cdot x\right)$$

Utilizing our homogeneous boundary conditions we get that:

$$\phi(0) = A\cdot 1 + B\cdot 0 = 0 \quad \therefore \quad A = 0$$

And,

$$\phi(\pi) = B\ \sin\left(\sqrt{\lambda+1}\cdot\pi\right) = 0$$

The above line implies:

$$\sqrt{\lambda+1}\cdot\pi = n\pi \ \rightarrow \quad \lambda_n = n^2 - 1$$

9.4: Fredholm Alternative and Generalized Green's Functions

Notice that $\lambda_n = n^2 - 1$ for n = 1,2,3,4… means that $\lambda = 0$ for n = 1; hence since there is an eigenvalue of '0' this means that there is a corresponding eigen-function $\phi(x) = \sin x$ (that is clearly non-trivial) that solves our homogeneous equation with the same boundary conditions.

Summary:

a) If $u_h(x) = 0$ is the only homogeneous solution → λ = 0 is NOT an eigenvalue ∴ the nonhomogeneous problem has a unique solution

b) $\phi_h(x) \neq 0$ → λ = 0 is an eigenvalue ∴ the nonhomogeneous problem has no solutions or infinite solutions

9.4: Fredholm Alternative and Generalized Green's Functions

So what does this ultimately mean for our formula for the coefficients:

$$-a_n\lambda_n = \frac{\int_0^L f(x)\phi_n(x)\,dx}{\int_0^L \phi_n^2(x)\,dx} \rightarrow \begin{cases} \int_0^L f(x)\phi_h(x)\,dx = 0 \\ \int_0^L f(x)\phi_h(x)\,dx \neq 0 \end{cases}$$

9.4: Fredholm Alternative and Generalized Green's Functions

In other words, for the nonhomogeneous problem, solutions exist only if the forcing function $f(x)$ is orthogonal to all homogeneous solutions:

$$\int_0^L f(x)\phi_h(x)\,dx = 0 \;\rightarrow u(x) \text{ exists for } \mathrm{L}[u] = f(x)$$

9.4: Fredholm Alternative and Generalized Green's Functions

Example 1: Suppose $u_{xx} = e^x$ with $u_x(0) = u_x(L) = 0$

The homogeneous problem is $u_{xx} = 0$, integrating twice yields $u(x) = Ax + B$. If we use the homogeneous boundary conditions we get, $u_x(0) = A = 0$ so $u(x) = B$. So our solution to the homogeneous $\phi_h(x)$ equation is a constant.

If we utilize the table from the previous slide $\int_0^L f(x)\phi_h(x)\,dx = \int_0^L e^x \cdot B\,dx \neq 0$ so there are no solutions to the nonhomogeneous problem.

In fact, if we integrate the non-homogeneous equation twice we get,

$$u(x) = e^x + Ax + B$$

But the first derivative $u_x = e^x + A$ and B.C. yield A = 0; therefore, $u_x > 0\ \forall x$

9.4: Fredholm Alternative and Generalized Green's Functions

Example 2: Let us consider the equation:

$$u_{xx} + \left(\frac{\pi}{L}\right)^2 u = \beta + x \text{ with } u(0) = u(L) = 0$$

The corresponding spatial homogeneous problem is $\phi''(x) + \left(\frac{\pi}{L}\right)^2 \phi(x) = 0.$

The characteristic equation is $r^2 + \left(\frac{\pi}{L}\right)^2 = 0$ with roots $r_{1,2} = \pm i\frac{\pi}{L}$.

Therefore, the solution for $\phi(x)$ is, $\phi(x) = A\cos\frac{\pi x}{L} + B\sin\frac{\pi x}{L}$. The first boundary condition yields A = 0 so $\phi(x) = B\sin\frac{\pi x}{L}$. The second boundary condition yields: $\phi(L) = B\sin\pi = B \cdot 0 = 0$ this implies that $|B| \geq 0$. So $\phi(x) = B\sin\frac{\pi x}{L}$ is a solution to the homogeneous problem with homogeneous boundary conditions.

9.4: Fredholm Alternative and Generalized Green's Functions

So we have obtained the homogeneous solution, but what about the nonhomogeneous solution(s)?

Recall, since we may have a non-trivial solution $\phi(x) = B \sin\frac{\pi x}{L}$ as long as B > 0.

Consider,

$$\int_0^L f(x)\phi(x)\,dx = \int_0^L (\beta + x)\, B \sin\frac{\pi x}{L}\,dx = 0$$

So solutions will occur as long as the forcing function is orthogonal to our spatial homogeneous solution. Therefore,

$$\beta = \frac{-\int_0^L x \sin\frac{\pi x}{L}\,dx}{\int_0^L \sin\frac{\pi x}{L}\,dx} = -\frac{L}{2}$$

9.4: Fredholm Alternative and Generalized Green's Functions

So how to we find the solution to $\mathrm{L}[u] = f(x)$ when $\lambda = 0$ is an eigenvalue, if a solution even exists of course???

Recall that if $\lambda = 0$ is NOT an eigenvalue, then we can utilize Green's function i.e.:

$$G(x, x_0) = \sum_{n=1}^{\infty} \frac{\phi_n(x)\phi_n(x_0)}{-\lambda_n \int_0^L \phi_n^2(x)\, dx}$$

In this case the function above satisfies:

$$\mathrm{L}[G(x, x_0)] = \delta(x - x_0)$$

9.4: Fredholm Alternative and Generalized Green's Functions

We will consider the case $\lambda = 0$ and $\phi_h(x) \neq 0$, the non-trivial case that solves the homogeneous equation $\mathrm{L}[\phi_h(x)] = 0$. Therefore, $\int_0^L f(x)\phi_h(x)\,dx = 0$.

But, notice that since $f(x) = \delta(x - x_0)$ the integral above:

$$\int_0^L \delta(x - x_0)\phi_h(x)\,dx = \phi_h(x_0) \neq 0$$

It would appear we have arrived at contradiction when we use Green's Function.

So let us introduce a similar problem instead, let:

$$\mathrm{L}[G(x, x_0)] = \delta(x - x_0) + c\phi_h(x)$$

9.4: Fredholm Alternative and Generalized Green's Functions

$$\mathrm{L}[G(x,x_0)] = \delta(x-x_0) + c\phi_h(x) = f(x)$$

Remember $\int_0^L f(x)\phi_h(x)\,dx = 0$; so we get,

$$0 = \int_0^L [\delta(x-x_0) + c\phi_h(x)]\phi_h(x)\,dx$$

$$0 = \int_0^L \delta(x-x_0)\phi_h(x)\,dx + c\int_0^L \phi_h^2(x)dx$$

$$0 = \phi_h(x_0) + c\int_0^L \phi_h^2(x)dx \;\rightarrow\; c = -\frac{\phi_h(x_0)}{\int_0^L \phi_h^2(x)dx}$$

We have successfully found a 'c' value expression and if we substitute it we will obtain our **Generalized Green's function.**

9.4: Fredholm Alternative and Generalized Green's Functions

$$\mathrm{L}[G(x,x_0)] = \delta(x-x_0) + c\phi_h(x) = f(x)$$

Becomes,

$$\mathrm{L}[G_m(x,x_0)] = \delta(x-x_0) - \frac{\phi_h(x_0)\phi_h(x)}{\int_0^L \phi_h^2(x)dx} = f(x)$$

Thus a particular solution to our $\mathrm{L}[u] = f(x)$ can be expressed as:

$$u(x) = \int_0^L f(x_0)G_m(x,x_0)dx_0$$

9.4: Fredholm Alternative and Generalized Green's Functions

Example: Suppose $u_{xx} = f(x)$ with B.C.: $u_x(0) = u_x(L) = 0$

The corresponding spatial homogeneous problem is $\phi''(x) = 0$ with $\phi'(0) = \phi'(L) = 0$.
Integrating twice yields $\phi_h(x) = Ax + B$. Next we consider the B.C.: $\phi_h'(x) = A \rightarrow \phi_h'(0) = A = 0$.
This means that $\phi_h(x) = B = constant$!

Therefore, for a solution to exist by Fredholm $\int_0^L f(x)\phi_h(x)dx = \int_0^L f(x) \cdot B \; dx = 0$.
We will consider f(x) to satisfy this.

Now let us try and find Green's function. Remember we utilized a comparison problem:

$$\mathrm{L}[G(x, x_0)] = \delta(x - x_0) + c\phi_h(x) \;\rightarrow\; \mathrm{L}[G_m(x, x_0)] = \delta(x - x_0) + c \cdot B$$

For the Green's function to exist the right hand side must be orthogonal to our homogeneous solution:

$$\int_0^L (\delta(x - x_0) + c \cdot B) \cdot B \; dx = 0$$

9.4: Fredholm Alternative and Generalized Green's Functions

For simplicity let B = 1, the previous integral becomes:

So, $\mathrm{L}[G_m(x, x_0)] = \delta(x - x_0) + c \cdot B$ becomes,

$$\frac{d^2 G_m}{dx^2} = \delta(x - x_0) - \frac{1}{L} = \begin{cases} -\frac{1}{L} & x \neq x_0 \\ \delta(0) - \frac{1}{L} & x = x_0 \end{cases}$$

Integrate the above line once to get

$$\frac{dG_m}{dx} = \begin{cases} -\frac{x}{L} & x < x_0 \\ 1 - \frac{x}{L} & x > x_0 \end{cases}$$

9.4: Fredholm Alternative and Generalized Green's Functions

Integrate once more to get the Green's function for this particular equation:

$$G_m(x, x_0) = \begin{cases} -\dfrac{x^2}{2L} + x_0 + c(x_0) & x < x_0 \\ x - \dfrac{x^2}{2L} + c(x_0) & x > x_0 \end{cases}$$

Therefore a particular solution to our equation is ,

$$u(x) = \int_0^L f(x_0) G_m(x, x_0) dx_0$$

9.5: Green's Functions for Poisson's Equation

We will analyze equations of the form:

$$\mathrm{L}[u] = f \text{ where } \mathrm{L} = \nabla^2$$

So the corresponding n-dimensional equations would look like this:

1-Dimensional: $u_{xx} = f(x)$

2-Dimensional: $u_{xx} + u_{yy} = f(x, y)$

3-Dimensional: $u_{xx} + u_{yy} + u_{zz} = f(x, y, z)$

N-Dimensional: $\sum_{k=1}^{N} u_{x_k x_k} = f(x_1, x_2, x_3 \ldots, x_k, \ldots x_N)$

9.5: Green's Functions for Poisson's Equation

We will use the following Green's Formulas for the Laplacian:

- 1-Dimensional: $\int u\nabla^2 v - v\nabla^2 u \, dx = uv_x - vu_x|_0^L$
- 2-Dimensional: $\iint u\nabla^2 v - v\nabla^2 u \, dA = \oint (u\nabla v - v\nabla u) \cdot \hat{\boldsymbol{n}} \, ds$
- 3-Dimensional: $\iiint u\nabla^2 v - v\nabla^2 u \, dV = \oiint (u\nabla v - v\nabla u) \cdot \hat{\boldsymbol{n}} \, dS$

Since we are dealing with variables in N-dimensional space; let,

$$\boldsymbol{x} = x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} \text{ and } \boldsymbol{x_0} = x_0\hat{\boldsymbol{\imath}} + y_0\hat{\boldsymbol{\jmath}}$$

$$f(\boldsymbol{x}) = f(x, y)$$

With, $\delta(\boldsymbol{x} - \boldsymbol{x_0}) = \delta(x - x_0)\delta(y - y_0)$ and the corresponding integral property:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\boldsymbol{x})\delta(\boldsymbol{x} - \boldsymbol{x_0}) \, dA = f(\boldsymbol{x_0})$$

9.5: Green's Functions for Poisson's Equation

Recall the 1-dimensional analysis for Green's Functions:

1-Dimensional

Given: $u_{xx} = f(x)$

Green's: $\frac{d^2G}{dx^2} = \delta(x - x_0)$

Solution: $u(x) = \int_0^L f(x_0)G(x, x_0)dx_0$

2-Dimensional

Given: $u_{xx} + u_{yy} = f(x, y)$

Green's: $\frac{d^2G}{dx^2} + \frac{d^2G}{dy^2} = \delta(x - x_0)\delta(y - y_0)$

Solution: $u(\boldsymbol{x}) = \iint f(\boldsymbol{x_0})G(\boldsymbol{x}, \boldsymbol{x_0})dA_0$

Recall from the 1-dimensional case (remember $\mathrm{L} = \frac{d^2}{dx^2} = \nabla^2$):

$$\int_0^L u\nabla^2 G - G\nabla^2 u \, dx = \int_0^L u(x)\delta(x - x_0) - G(x, x_0)f(x) \, dx = 0$$

9.5: Green's Functions for Poisson's Equation

The 2-dimensional case is analogous,

$$\iint u(\boldsymbol{x})\nabla^2 G(\boldsymbol{x},\boldsymbol{x_0}) - G(\boldsymbol{x},\boldsymbol{x_0})\nabla^2 u(\boldsymbol{x})\, dA = 0 \rightarrow \iint u(\boldsymbol{x})\delta(\boldsymbol{x}-\boldsymbol{x_0}) - G(\boldsymbol{x},\boldsymbol{x_0})f(\boldsymbol{x})\, dA = 0$$

If we separate the last integral above we get:

$$\iint u(\boldsymbol{x})\delta(\boldsymbol{x}-\boldsymbol{x_0})\, dA - \iint G(\boldsymbol{x},\boldsymbol{x_0})f(\boldsymbol{x})\, dA = 0$$

If interchange the roles of $\boldsymbol{x}, \boldsymbol{x_0}$ we get: $u(\boldsymbol{x}) = \iint G(\boldsymbol{x},\boldsymbol{x_0})f(\boldsymbol{x_0})\, dA_0$

9.5: Green's Functions for Poisson's Equation

Let us now use eigenfunction expansions and the Fredholm Alternative to solve Poisson's Equation:

$$\mathrm{L}[u] = \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f(x, y, z) = f(\boldsymbol{x})$$

Recall, that our solution *u* can be expanded into a series using the eigenfunctions:

$$u(\boldsymbol{x}) = \sum_{n=1}^{\infty} a_n \phi_n(\boldsymbol{x})$$

Since we can differentiate term by term:

$$f(\boldsymbol{x}) = \nabla^2 u = \sum_{n=1}^{\infty} a_n \nabla^2 \phi_n(\boldsymbol{x}) = -\sum_{n=1}^{\infty} \lambda_n a_n \phi_n(\boldsymbol{x})$$

9.5: Green's Functions for Poisson's Equation

$$f(\boldsymbol{x}) = \nabla^2 u = \sum_{n=1}^{\infty} a_n \nabla^2 \phi_n(\boldsymbol{x}) = -\sum_{n=1}^{\infty} \lambda_n a_n \phi_n(\boldsymbol{x})$$

Therefore, we can find the coefficients $\lambda_n a_n$:

$$-\lambda_n a_n = \frac{\iint f(\boldsymbol{x_0})\phi_n(\boldsymbol{x_0})dA_0}{\iint \phi_n(\boldsymbol{x})dA} \rightarrow a_n = \frac{\iint f(\boldsymbol{x_0})\phi_n(\boldsymbol{x_0})dA_0}{-\lambda_n \iint \phi_n(\boldsymbol{x})dA}$$

Substitute into $u(\boldsymbol{x}) = \sum_{n=1}^{\infty} a_n \phi_n(\boldsymbol{x})$ we get,

$$u(\boldsymbol{x}) = \sum_{n=1}^{\infty} \frac{\iint f(\boldsymbol{x_0})\phi_n(\boldsymbol{x_0})dA_0}{-\lambda_n \iint \phi_n(\boldsymbol{x})dA} \phi_n(\boldsymbol{x}) = \iint f(\boldsymbol{x_0}) \sum_{n=1}^{\infty} \frac{\phi_n(\boldsymbol{x_0})\phi_n(\boldsymbol{x})}{-\lambda_n \iint \phi_n(\boldsymbol{x})dA} dA_0$$

9.5: Green's Functions for Poisson's Equation

We have found our Green's Function:

$$G(\boldsymbol{x}, \boldsymbol{x_0}) = \sum_{n=1}^{\infty} \frac{\phi_n(\boldsymbol{x_0})\phi_n(\boldsymbol{x})}{-\lambda_n \iint \phi_n(\boldsymbol{x}) dA}$$

Again, what happens when λ = 0? We will refer back to the Fredholm Alternative:

- If λ ≠ 0, then there is a unique solution to the non-homogenous solution.
- If λ = 0, and $\iint f(\boldsymbol{x_0})\phi_h(\boldsymbol{x_0}) dA_0 = 0$, then there are infinite solutions.
- If λ = 0, and $\iint f(\boldsymbol{x_0})\phi_h(\boldsymbol{x_0}) dA_0 \neq 0$, then there are no solutions solutions.